Lab 8

H-Bridge Motor Controller

Milestone

Goal

Build the Verilog (FPGA) and Python (host) needed to control a DC gearmotor through the Pmod DHB1 while streaming acceleration data from the on-board LSM303. Reuse your LSM303 readout framework from the previous lab and add motor control. Then use the accelerometer on the tilted rig to measure the arm's angle, and protect the motor from a stall.

Hardware setup

The rig

- The sensor board assembly is the arm, mounted on the gearmotor's output shaft: about 300 g, roughly balanced, with the IMU about 10 cm from the shaft.
- The arm travels about 300° between solid hard stops (the post).
- Put a book under one side of the base so the rotation axis tilts about 15° from vertical.
- Home the arm at the centre of its travel, so its angle runs from about −150° to +150° and atan2 never wraps.

The driver

- Plug the Pmod DHB1 into the sensor board's PMOD header.
- Power: connect two wires from the bench supply to J4 on the DHB1 (VMOTOR and GND). Double-check polarity on both the supply and the Pmod. Keep VMOTOR at or below 6 V, the motor's rating.
- Watch current limits for the DHB1/DRV8833 (1.5 A RMS per bridge) and the motor (1.8 A at stall).
- The DC motor uses one H-bridge, motor 1: EN1 takes the pulses (PWM) and DIR1 sets the direction. Motor 2 is not used.
- Safety: keep leads short and secure; verify no shorts before enabling power.

> **The motor will not notice hitting the post.** Stalled, it draws about 1.8 A (about 11 W at 6 V), and nothing in the driver is guaranteed to stop it. It simply heats. Your code is the protection.

Functional requirements

1. **Host control → FPGA**
   - The user enters direction and pulse count on the PC.
   - Send these settings to the FPGA over USB (OK Wires or BTPipes).
2. **Pulse generation on FPGA**
   - Generate a 200 Hz pulse train on EN1 (PWM); DIR1 selects the direction.
   - Output exactly the requested number of pulses for the selected direction.
   - When the pulses complete, hold EN1 low to stop the motor.
   - Change DIR1 only while EN1 is low.

- For a DC motor, the pulse count sets how long the motor runs (N ÷ 200 s), not a fixed angle.

3. **Concurrent sensor acquisition**
   - Read the accelerometer (X/Y/Z) continuously, also while the motor spins, and stream the readings to the PC.
   - Sampling must be independent of the pulses (free-running task, not one sample per pulse).
4. **Tilt angle**
   - With the arm stopped, measure the tilt β from the two accelerometer axes in the arm's plane (radial, along the arm; tangential, through the board face):

$$\sqrt{(a_{radial}^2 + a_{tangential}^2)} = g \cdot \sin \beta$$

5. **Arm angle, read only when stopped**
   - Compute the arm angle $\phi = \text{atan2}(-a_{tangential}, a_{radial})$.
   - Use this cycle for every angle measurement: command the move → stop → wait for the arm to settle → read.
   - While the arm turns, centripetal acceleration $\omega^2 r$ adds to the radial axis, so an angle read while moving is wrong.
6. **Resolution**
   - At one position, take 100 readings of ϕ without moving. Their standard deviation (in degrees) is the resolution.
   - Compare it with what you predict from β and the accelerometer noise.
7. **Stall protection**
   - Detect a stall: EN1 is driving the motor but the angle stops changing.
   - On a stall, drive EN1 low (safe state) and log what the detector saw and when it tripped.
   - With protection running, command the arm into the post on purpose.

What the TA checks

#	The TA does this	This must happen
1	Asks for your measured tilt angle	a number, and it is near 15 degrees
3	Asks for the standard deviation of 100 stationary readings	a number in degrees, and it is close to what you predicted
4	Asks when you read the accelerometer	only when stopped, and you can say why
8	Asks you to drive into the post	the detector trips, the bridge goes to your chosen safe state, and the log shows both

Check 8 is the one that cannot be faked afterwards. Have the rig positioned so the arm has somewhere safe to jam.

From the pre lab

Predicted resolution (check 3, question 1(a))

$$\delta\phi = 1\ \text{mg} \div (g \cdot \sin\beta) = 0.001 \div \sin 15° = 0.001 \div 0.259 = 0.00386\ \text{rad} \approx 0.22°$$

Moment of inertia (question 4(a))

$$J = m \cdot L^2 / 12 = 0.3\ \text{kg} \times (0.28\ \text{m})^2 / 12 = 1.96 \times 10^{-3}\ \text{kg}\cdot\text{m}^2$$

Stall detector and safe state (check 8)

```
WINDOW   = 100 ms
EXPECTED = normal move per WINDOW at your VMOTOR   # measure it in a free run
MIN_FRAC = 0.25       # trip if moved < 25% of EXPECTED
STRIKES  = 2          # stall lasts ≤ 200 ms

every WINDOW while EN1 is pulsing:
    phi_now = atan2(−a_tangential, a_radial)     # average 4–8 samples
    moved   = unwrap(phi_now − phi_prev)

    if sign(moved) != dir or |moved| < MIN_FRAC ∗ EXPECTED:
        strikes += 1
    else:
        strikes = 0

    if strikes >= STRIKES:
        stop the pulses, EN1 = 0
        latch fault, log time, phi, pulses sent, EN1 → 0
        no moves until reset and re−home

    phi_prev = phi_now
```

- A DC motor has no steps, so EXPECTED comes from a measurement, not steps × 1.8°. For scale: 150 rpm (no load, 6 V) is 90° per 100 ms; under load and with 200 Hz pulses it is less.
- At a stall ω = 0, so the readings the detector trips on have no $\omega^2 r$ error.
- Safe state: disabled and coasting (EN1 low). Driving into a jam keeps about 1.8 A (about 11 W at 6 V) in a motor that isn't turning, and the arm already rests against the post. The DHB1 has no brake; EN low always coasts.

Post lab questions

Question 1(a): How good is your protractor?

Report the standard deviation of your 100 stationary readings, in degrees. Compare it with the resolution you predicted in the pre lab, and account for any difference.

Question 3(b): Ringing

From a continuous accelerometer log of the arm coming to a stop, report the overshoot and the settling time. Then turn that into a rule: what is the fastest rate at which this rig can make trustworthy stop-and-

read position measurements? Give a number in measurements per second.

> Question 3(b) is the one that matters outside this lab. A settling time you have measured is a limit on your whole system, and it was invisible until you stopped reading only when things were still.

Question 4(a): Torque, with no torque sensor

From a continuous accelerometer log of the arm speeding up from rest, extract the angular acceleration α from the tangential axis. Using the J you computed in the pre lab, report the torque the motor actually produced, in N·m ($\tau = J \cdot \alpha$).

Question 4(c): Torque at two voltages

Repeat 4(a) at two supply voltages. Report both torques and say whether the difference is what you expect, and why.